Figure 4: Installing OpenSSH using Cydia

OpenSSH (Figure 4). This will help in connecting to iPhone over Wi-Fi and to run UNIX commands. If you want, you can also install an application called Terminal, which will give you a shell environment on your iPhone. The default shell for iPhone is Bash.

Also note down the IP address of your iPhone. Connect your iPhone to the Wi-Fi network. Go to *Settings→Wifi→ Wifi Acess Point Name*, and note down the IP address.

Building the free tool chain

In this section, we will build the tool chain for iPhone, which includes the compiler, linker, etc. I am using OpenSUSE 11.1, but this should work on any Linux distro.

Make sure you install the following software on your Linux distribution before proceeding:

1. Basic development tools including GCC, Make, Autotools, Flex, Bison, autoconf, automake, etc.
2. Objective C Support for GCC is very important. Search for Object C or Objective C in your package manager
3. Git and SVN
4. cpio
5. libxml2-dev, needed to build xar
6. xar (*code.google.com/p/xar*) and *dmg2img* [available on *LFY DVD/software/magazine/iPhone*). Build and install xar:

```
$ tar xvfz xar*
$ cd xar*
$ ./configure –prefix=/usr && make
$ sudo make install
```

   Build and install dmg2img:

```
$ tar xvfz dmg2img*
$ make
$ sudo cp dmg2img /usr/bin/dmg2img
```

7. MacOSX10.4u.sdk—This one is little tricky. Here is how to get it:
   - Download "Xcode for iPhone and Mac Development" from *developer.apple.com/technology/Xcode.html.*
   - Convert the dmg file to a flat img file

```
$ dmg2img -i iphonesdk.dmg -o iphonesdk.img -v
```

   - Mount the img file to */tmp/iphonesdk*. Make sure you are logged in as the root. Or use the *sudo* command.

```
# modprobe hfsplus
# mount -t hfsplus -o loop iphonesdk.img /tmp/iphonesdk
```

   - Copy the MacOSX10.4.Universal.pkg

```
# mkdir /tmp/extract
# cp Packages/MacOSX10.4.Universal.pkg /tmp/extract
```

   - Extract the pkg file using xar

```
# cd /tmp/extract
# xar -x -v -f *.pkg
```

   - You will have a big file called Payload. Extract the payload file

```
# mv Payload Payload.gz
# gunzip Payload.gz
# cat Payload | cpio –i
```

   - Finally copy the folder MacOSX10.4u.sdk

```
# cd SDK
# mkdir /isdk
# cp -r MacOSX10.4u.sdk /isdk/
```

What's this 'Objective-C' thing?

The Wikipedia says: "Objective-C is a reflective, object-oriented programming language, which adds Smalltalk-style messaging to C. Today it is used primarily on Apple's Mac OS X and iPhone OS, two environments based on the OpenStep standard, and it is the primary language used for Apple's Cocoa API, though it was originally used as the main language on NeXT's NeXTSTEP OS. Generic Objective-C programs, which do not make use of these libraries, can also be compiled for any system supported by gcc, which includes an Objective-C compiler." For more information refer to *en.wikipedia.org/wiki/Objective-C.*

This essentially means that if you want to program for MAC OS X and iPhone, you have to do it in Objective C. Fortunately, all Linux distributions support Objective C.

Copy the iPhone file system

We will need to copy the iPhone file system that will help us in getting iPhone libraries and frameworks.

Warning: I am not a lawyer. Check Apple's *Terms and Conditions* yourselves to make sure that you are not breaking any laws.

Do the following:

```
# mkdir -p /toolchain/sys/
# cd /toolchain/sys/
# mkdir -p ./System/Library ./usr
# scp -r root@<iPhone IP Address>:/System/Library/Frameworks/ ./System/
Library
# scp -r root@<iPhone IP Address>:/:/System/Library/PrivateFrameworks/
./System/Library
# scp -r root@<iPhone IP Address>:/:/usr/lib ./usr
```